

Can nutrition limit exercise-induced immunodepression?

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Prolonged exercise and heavy training are associated with depressed immune cell function. To maintain immune function, athletes should eat a well-balanced diet sufficient to meet their energy, carbohydrate, protein, and micronutrient requirements. Consuming carbohydrate during prolonged strenuous exercise attenuates rises in stress hormones and appears to limit the degree of exercise-induced immune depression. Recent evidence suggests that antioxidant vitamin supplementation may also reduce exercise stress and impairment of leukocyte functions. Further research is needed to evaluate the effects of other antioxidants and dietary immunostimulants such as probiotics and echinacea on exercise-induced immune impairment.